10.14 Dimensional Analysis with $\hbar$, and What $[g] = 0$ Means

LecNote11 pp. 1–2 set $\hbar = c = 1$, give every quantity a mass dimension, and sort the couplings g_n of $\frac{1}{n!} g_n \varphi^n$ by the sign of $[g_n]$. Three questions are left over:

- "The title on p. 1 says 'Naïve'. What does $\hbar = 1$ hide, and how do we put $\hbar$ back?"
- "For $[g] < 0$ the argument of f blows up (p. 2). So what? Why is Srednicki's φ^3 in $d = 6$?"
- "For $[g] = 0$ the argument is g at every energy. Isn't that boring? Why 'fun across scales'?"

Conventions are Srednicki's: metric $(-,+,+,+)$, $k^2 = -m^2$ on shell, $\tilde{\Delta}(k) = 1/(k^2 + m^2 - i\epsilon)$, $\langle 0|T\varphi(x)\varphi(y)|0\rangle = \frac{1}{i}\Delta(x-y)$ in the free theory, $s = -(k_1 + k_2)^2$; in φ^3 theory $\mathcal{L}_1 = \frac{1}{6} Z_g g \varphi^3 + \mathcal{L}_{\text{ct}}$, $\mathcal{L}_{\text{ct}} = -\frac{1}{2} A\, \partial\varphi\partial\varphi - \frac{1}{2} B m^2 \varphi^2 + Y\varphi$, $A = Z_\varphi - 1$, $B = Z_m - 1$, and $\alpha = g^2/(4\pi)^3$ (LecNote12 p. 11). $[X]$ is the mass dimension of X at $\hbar = c = 1$.

Units, and $\hbar$ put back

Since $[\hbar] = E \cdot T$ and $[c] = L/T$, setting both to 1 makes times and lengths inverse energies; $\hbar c = 197.327$ MeV fm $= 0.197327$ eV μm (p. 1 rounds it to 0.2) and $\hbar = 6.5821 \times 10^{-22}$ MeV s convert back: $1/(1\text{ GeV})$ is 0.1973 fm, or 6.58×10^{-25} s.

To see what $\hbar = 1$ hides, keep $c = 1$ but let $\hbar$ keep its units, mass times length. The action has the units of $\hbar$, so the kinetic term gives φ the units of $\hbar^{1/2}(\text{length})^{1-d/2}$, the m in $\mathcal{L}$ is an inverse length (the mass is $\hbar m$), and $g_n \varphi^n$ gives g_n the units of $\hbar^{1-n/2}(\text{length})^{-[g_n]}$. Hence

$$g_n \hbar^{(n-2)/2} \text{ has the units of } (\text{length})^{-[g_n]}, \tag{10.14.1}$$

and when $[g_n] = 0$ the pure number is $g^2\hbar$ for φ^3 in $d = 6$ ($\lambda\hbar$ for φ^4 in $d = 4$), not g alone.

Loops count powers of $\hbar$

With $\hbar$ kept, $Z(J) = \int \mathcal{D}\varphi \, \exp\left[\frac{i}{\hbar}\int d^d x\, (\mathcal{L} + J\varphi)\right]$. At $g = 0$ it is $\exp\left[\frac{i}{2\hbar}\int J\Delta J\right]$, so a propagator is $\frac{\hbar}{i}\Delta$, a vertex $ig/\hbar$, a source $iJ/\hbar$. A connected diagram with P propagators, V vertices and E sources goes as $\hbar^{P-V-E}$; counting the sources as endpoints, it has $L = P - V - E + 1$ loops. So

$$\text{diagram} \propto \hbar^{L-1}, \qquad Z = e^{\frac{i}{\hbar}W}, \quad W = W_0 + \hbar W_1 + \hbar^2 W_2 + \cdots, \tag{10.14.2}$$

and a connected E-point function, whose E source factors $1/\hbar$ are stripped off, goes as $\hbar^{L+E-1}$ at L loops. Two consequences. *Tree level is classical*: as $\hbar \to 0$ the stationary point $\delta S/\delta\varphi = -J$ dominates and $W_0 = S[\varphi_{\rm cl}] + \int J\varphi_{\rm cl}$; each loop is one more power of $\hbar$. *Each loop costs* $g^2\hbar$: at fixed E a φ^3 diagram has $V = E - 2 + 2L$ (§10.11), so the L-loop term carries $g^{E-2}\hbar^{E-1}(g^2\hbar)^L$: the expansions in $\hbar$ and in g^2 are one expansion, in $g^2\hbar$, the pure number of (10.14.1) at $d = 6$.

A zero-dimensional check: with the weight $\exp[-(\frac{1}{2}\varphi^2 - \frac{g}{6}\varphi^3 - J\varphi)/\hbar]$, as a formal series in g (the integral itself diverges; §10.17), the connected moments $\langle\varphi^E\rangle_C = \hbar^E \partial_J^E \ln Z|_{J=0}$ are

$$\begin{aligned} \langle\varphi\rangle_C &= \tfrac{1}{2}g\hbar + \tfrac{5}{8}g^3\hbar^2 + \tfrac{15}{8}g^5\hbar^3 + \cdots, \\ \langle\varphi^2\rangle_C &= \hbar + g^2\hbar^2 + \tfrac{25}{8}g^4\hbar^3 + \cdots, \\ \langle\varphi^3\rangle_C &= g\hbar^2 + 4g^3\hbar^3 + \tfrac{175}{8}g^5\hbar^4 + \cdots, \end{aligned} \tag{10.14.3}$$

each term $g^{E-2+2L}\hbar^{L+E-1}$, the L-loop power ($\langle\varphi\rangle$ has no tree term because $\varphi_{\rm cl} = 0$ at $J = 0$), and the $\hbar^0$ part of $\hbar \ln Z(J)$ is $J\varphi_{\rm cl} - S(\varphi_{\rm cl})$ with $S'(\varphi_{\rm cl}) = J$, the classical limit.

What the sign of $[g]$ means

Amplitudes. By $\langle k|k'\rangle = (2\pi)^{d-1}2\omega\,\delta^{d-1}(\mathbf{k} - \mathbf{k}')$ a one-particle state has dimension $-(d-2)/2 = -[\varphi]$. With n particles in and out together, stripping $(2\pi)^d\delta^d$ from $\langle f|i\rangle = (2\pi)^d\delta^d(\cdots)\,iT_n$ leaves $[T_n] = d - \frac{n}{2}(d-2) = [g_n]$, as a single φ^n vertex, $T_n \propto g_n$, requires. If the masses drop out at $s \gg m^2$ (but see the last subsection), one coupling leaves only angles and $g\,s^{-[g]/2}$ as pure numbers:

$$T_n = s^{[T_n]/2}\,F\big(g\,s^{-[g]/2},\ \text{angles}\big). \tag{10.14.4}$$

This is p. 2's argument, with $\bar g \equiv g/\Lambda^{[g]}$ so that $\bar g\,(\Lambda^2/s)^{[g]/2} = g\,s^{-[g]/2}$ (the page's $(g/\Lambda)^{[g]}$ has dimension $[g]([g]-1)$). In any d the φ^3 tree amplitude (§10.11) at $s, |t|, |u| \gg m^2$ is $T = -g^2(1/s + 1/t + 1/u) = s^{(4-d)/2}(g\,s^{-[g]/2})^2 F(t/s, u/s)$: its strength $g^2 s^{(d-6)/2}$ falls like $1/s$ in $d = 4$, is constant in $d = 6$, and grows like s in $d = 8$. In general $g\,s^{-[g]/2} = 1$ at $s = g^{2/[g]}$; above that scale the interaction fades for $[g] > 0$, and the expansion in g fails for $[g] < 0$.

Power counting. A diagram for T_E is a product of couplings times a loop integral $\int (d^dl)^L/(l^2)^I$ of dimension $\omega = dL - 2I$, its superficial degree of divergence. Matching dimensions (or using $\sum_n nV_n = 2I + E$ and $L = I - \sum_n V_n + 1$),

$$\omega = [T_E] - \sum_n V_n[g_n] = d - \tfrac{E}{2}(d-2) - \sum_n V_n[g_n]. \tag{10.14.5}$$

So the sign of $[g]$ decides how ω changes with the order (LecNote14, §18):

- $[g] > 0$, *super-renormalizable*: ω falls as V grows, so only finitely many diagrams are superficially divergent (φ^3 in $d = 4$: $\omega = 4 - E - V$).
- $[g] = 0$, *renormalizable*: $\omega = d - \frac{E}{2}(d-2)$ at every order, so the same few E diverge, with counterterms already in $\mathcal{L}$: in φ^3, $d = 6$, $\omega = 4, 2, 0$ for $E = 1, 2, 3$ (Y; A and B; Z_g).
- $[g] < 0$, *non-renormalizable*: ω grows with V, so every E eventually needs a counterterm: in $d = 7$ the two-loop φ^3 four-point function ($V = 6$, $I = 7$) has $\omega = 0$ and needs a φ^4 one.

Hence Srednicki's $d = 6$: $[g_3] = (6-d)/2$ vanishes only there, and φ^3 is non-renormalizable for every $d > 6$; LecNote12 p. 7's $d < 8$ bounds only the one-loop Π, whose Π'' has $\omega = d - 8$.

$[g] = 0$: LOGARITHMS

If the masses simply dropped out at $s \gg m^2$, a theory with $[g] = 0$ would look the same at every energy. They do not. LecNote12's on-shell self-energy (p. 13), $\Pi = \frac{1}{2}\alpha \int_0^1 dx\, D \ln(D/D_0) - \frac{\alpha}{12}(k^2 + m^2)$ with $D = x(1-x)k^2 + m^2$, $D_0 = [1 - x(1-x)]m^2$, behaves at large spacelike k^2 as

$$\frac{\Pi(k^2)}{k^2 + m^2} \longrightarrow \frac{\alpha}{12}\Big[\ln\frac{k^2}{m^2} + 3 - \pi\sqrt{3}\Big]. \tag{10.14.6}$$

The constant is p. 14's c_1, called C_1 on p. 15. The logarithm has no $m \to 0$ limit. In the exact propagator $\tilde{\mathbf{\Delta}} = [1 - \Pi/(k^2 + m^2)]^{-1}/(k^2 + m^2)$ it is a correction $(\alpha/12)\ln(k^2/m^2)$ relative to the tree term, as large as the tree term at $k^2 \sim m^2 e^{12/\alpha}$ however small α is: fixed-order perturbation theory then fails because of the logarithm, not the coupling. With $[g] = 0$ no power of s competes with it: this is the "fun across scales". Organizing these logarithms, with a coupling that then depends on the energy, is the renormalization group (LecNote15, §§26–28; §10.19).

SUMMARY

Diagrams go as $\hbar^{L-1}$: tree level is classical; each φ^3 loop costs $g^2\hbar$. Couplings enter as $g\, s^{-[g]/2}$; the sign of $[g]$ is the class, and $[g_3] = 0$ only at $d = 6$. At $[g] = 0$ loops bring $\ln(k^2/m^2)$.